

Coding Exercise 3

1. (a) Write a code to generate an $MA(q)$ random process up to $t = 10000$ for a given value of q and $a_1, a_2, \dots, a_q$. Consider the process $\{\varepsilon_1, \varepsilon_2, \dots\}$ as i.i.d. random variables generated according to some distribution whose mean is 0.5 and variance is 2.
(b) Verify if the empirical mean and variance of the random variables in $MA(q)$ are close to the theoretical mean and variance. Do the same for the autocovariance values $\gamma(1), \gamma(2), \dots, \gamma(q+2)$.
2. Repeat the above problem for an $AR(3)$ process for a given value of a_1, a_2, a_3 . To verify autocovariance, compute the theoretical values of $\gamma(0), \dots, \gamma(5)$ using Yule-Walker equations.
3. The first column of *bjm.csv* file has been generated using an $AR(p)$ process. Compute the mean and variance of the underlying i.i.d. process. Also compute the value of p and the coefficients $a_1, a_2, \dots, a_p$ using Yule-Walker equations.
4. The second and third columns of *bjm.csv* file have been generated using $ARMA(p_i, q_i)$ processes, $i = 2, 3$. Compute the mean and variance of the underlying i.i.d. process, the values of p_i and q_i , and the corresponding coefficients for the respective columns, using Box-Jenkins method.
5. (a) Write a code to generate a $VAR(p)$ random process up to $t = 10000$ for a given value of p and matrices $A_1, A_2, \dots, A_p$. Consider the process $\{\varepsilon_1, \varepsilon_2, \dots\}$ as i.i.d. random variables generated according to some joint distribution whose mean is $[0.5, -0.5]$ and covariance matrix is $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$.
(b) Assume that the generated data were given to you with a mention that it is a VAR process. Write a code to find p and the corresponding matrices $A_1, A_2, \dots, A_p$.